

Seunghwan Um

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RESEARCH STATEMENT

I'm a PhD student at [Robotics Innovatory](#), Sungkyunkwan University, South Korea, under the supervision of [Prof. Hyouk Ryeol Choi](#).

My research focuses on *robotic manipulation* for logistics applications, specifically picking and placing tasks. This work spans the entire spectrum, from initial gripper design to the current development of an imitation learning pipeline. Ultimately, to achieve human-level manipulation, I am deeply interested in integrating fingertip sensors and developing learning policies that can effectively leverage them.

EDUCATION

- **Sungkyunkwan University (SKKU)** Mar. 2022 - Present
Ph.D in Mechanical Engineering Suwon, Republic of Korea
- **Hanyang University (ERICA)** Mar. 2018 - Feb. 2022
B.S in Mechanical Engineering, Degree Honors: Cum Laude Ansan, Republic of Korea

PUBLICATIONS

J=JOURNAL, C=CONFERENCE, S=IN SUBMISSION, P=PATENT, T=THESIS

- [J.6] **PALM-Gripper: Integrated Gripper with Parallel Adaptable Mechanism for Shelf Picking in Logistics**
Seunghwan Um, Yeong Gwang Son, Juyong Hong, Chun Soo Kim, et al. and Hyouk Ryeol Choi*.
IEEE/ASME Transactions on Mechatronics (TMECH), 2026.
- [J.5] **Toward Reliable Bin-Picking: Collision-Aware Robotic Design and Control Strategy for Heavily Cluttered Environment**
Seunghwan Um, Yeong Gwang Son, Jaeyoon Shim, Hyouk Ryeol Choi*
IEEE Robotics and Automation Practice (RA-P), 2026, The paper was invited for publication.
- [J.4] **Plug-and-Play Shape Matching Module for Zero-Shot Mesh-Free Grasp Refinement on Unknown Objects**
Juyong Hong, Yeong Gwang Son, Seunghwan Um, Hyouk Ryeol Choi*.
IEEE Robotics and Automation Letters (RA-L), 2025.
- [J.3] **Corner-Grasp: Multi-Action Grasp Detection and Active Gripper Adaptation for Grasping in Cluttered Environments**
Yeong Gwang Son, Seunghwan Um, Juyong Hong, Tat Hieu Bui, Hyouk Ryeol Choi*.
ArXiv, 2025.
- [J.2] **Development of Adaptive Gripper Enhancing Power Grasp Range and Linearity**
Issac Rhee, Chun Soo Kim, Heeyeon Jeong, Seunghwan Um, and Hyouk Ryeol Choi* et al.
IEEE Access, 2024.
- [J.1] **ReC-Gripper: A Reconfigurable Combined Suction and Fingered Gripper for Various Logistics Picking and Stowing Tasks**
Seunghwan Um, Heeyeon Jeong, Chun Soo Kim, Issac Rhee, and Hyouk Ryeol Choi*
IEEE Robotics and Automation Letters (RA-L), Presented in ICRA 2024.
- [C.1] **Overcoming Heavy Clutter: Utilizing the Hybrid Grasping Network and Gripper**
Seunghwan Um, Yeong Gwang Son, Tat Hieu Bui, Ho Sang Jung, and Hyouk Ryeol Choi*
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024
Workshop: Benchmarking via Competitions in Robotic Grasping and Manipulation 🏆 [Best Extended Abstract]

PATENTS

- [P.6] **Horizontal Adaptive Hybrid Gripper and Robot Including the Same.**
Seunghwan Um, Chun Soo Kim, Issac Rhee, Heeyeon Jeong, and Yoon Haeng Lee.
Korean Intellectual Property Office, Application No. 10-2024-0131112. Application Date: 2024.09.26.
- [P.5] **Gripper Including Multi-Stage Translational Mechanism.**
Chun Soo Kim, Issac Rhee, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.
Korean Intellectual Property Office, Application No. 10-2023-0125840. Application Date: 2023.09.20.
- [P.4] **Gripper Including Telescopic Mechanism.**
Chun Soo Kim, Issac Rhee, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.
Korean Intellectual Property Office, Application No. 10-2023-0124012. Application Date: 2023.09.18.

- [P.3] **Suction Gripper Capable of Translational and Rotational Movements.**
Chun Soo Kim, Issac Rhee, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.
Korean Intellectual Property Office, Patent No. 10-2023-0093340. Registration Date: 2023.10.18.
- [P.2] **Adaptive Gripper Capable of Parallel Motion.**
Issac Rhee, Chun Soo Kim, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.
Korean Intellectual Property Office, Patent No. 10-2023-0077512. Registration Date: 2023.12.06.
- [P.1] **Hybrid Gripper Capable of Bin Picking and Shelf Picking.**
Seunghwan Um, Heeyeon Jeong, Chun Soo Kim, Issac Rhee, and Yoon Haeng Lee.
Korean Intellectual Property Office, Patent No. 10-2023-0076248. Publication Date: 2024.12.23.

HONORS AND AWARDS

- 🏆 **Robot Engineering and Design Show (RED Show)** Feb. 2026
Korea Robotics Society Annual Conference (KRoC) [KRoC RED Show 🌐]
- 🏆 **Samsung Humantech Paper Award** Feb. 2025
Samsung Electronics Co., Ltd. [Samsung Humantech 🌐]
- 🏆 **9th Robotic Grasping of Manipulation Competition - Picking in Clutter** May. 2024
IEEE, IEEE RAS [RGMC 2024 🌐]
- 🏆 **KSME Student Creative Design Competition** Oct. 2021
The Korean Society of Mechanical Engineers (KSME) [YouTube 🌐]
- 🏆 **Creative and Intelligent Robot Contest** Sep. 2021
Daejeon Metropolitan City, Chungnam National University [YouTube 🌐]

EXPERIENCE

- **CarbonSix** Oct. 2024 - Nov. 2024, Jun. 2026 - Jul. 2026
Researcher Gangnam-daero, Seocho-gu, Seoul, Republic of Korea
◦ Designed direct teaching device with gravity compensation function, and camera calibration device.
- **AIDIN ROBOTICS - Cobot Solution Team** May 2022 - Present
Researcher Anyang-si, Gyeonggi-do, Republic of Korea
◦ Designed grippers for shelf-picking solutions, contributing to efficient logistics automation.
- **Korea Institute of Industrial Technology (KITECH)** Oct 2021 - Dec 2021
Research Student Ansan-si, Gyeonggi-do, Republic of Korea
◦ Designed and developed control systems for a 2-DoF manipulator, enhancing its performance and accuracy.
- **Wall Climbing Car (WCC) [Undergraduate Project]** Dec 2020 - Oct 2021
Team Leader Ansan-si, Gyeonggi-do, Republic of Korea
◦ Designed a wall-climbing car's propeller frame and control system.

PERSONAL PROJECTS

- **Scooping Grasping with Robotic Hand | Wrist F/T sensing, Compliance control** May 2026 - Present
[🌐] [🔗]
◦ **Objective:** Enabling a rigid-fingertip robotic hand to scoop objects off the floor, a motion in which environmental contact would otherwise close a kinematic chain and damage the finger mechanism.
◦ Wrench measured in wrist is fed into a Cartesian impedance controller so that the manipulator responds compliantly to both wrist- and fingertip-level contacts, achieving safe interaction against external constraints without sacrificing the grasping force and positional accuracy.
- **Impedance Control with Doosan M0609** Jan. 2026 - Present
[🌐] [🔗]
◦ **Objective:** Overcoming the limitations of position control in teleoperated manipulation, where contact-rich tasks and constrained environments cannot be handled reliably.
◦ Implemented a task-space impedance controller running on a 1 kHz real-time torque-control loop over the Doosan real-time API.
◦ Developed a momentum-based disturbance observer (De Luca formulation) for external-contact estimation, together with gravity, tool, and friction compensation.
◦ Demonstrated contact maintenance under continuous force and impact, and pivoting tasks in constrained environments, where position control fails.

PROJECTS

- **Development of a K-Logistics Humanoid Robot Integrated with a High-Sensitivity Robotic Hand Based on a Multimodal AI Foundation Model** *Sep 2025 - Present*
Participating organizations: AIDIN ROBOTICS, SKKU, KETI, CJ Logistics Republic of Korea
 - **Research Objective:** Automation of contact-rich manipulation tasks using a humanoid robot equipped with a high-sensitivity robotic hand.
 - Development of a teleoperation system for constructing multimodal datasets incorporating force information
 - Design of force-control strategies and learning-based policies capable of handling contact-rich interactions
- **Development of Smart Vision System and All-in-One Universal Gripper for Multi-Variou Random Piece Picking** *May 2022 - 2024*
Participating organizations: SKKU, AIDIN ROBOTICS, KITECH, CJ Logistics Republic of Korea
 - **Research Objective:** Developing robotic picking system including a gripper and vision system for piece picking in a logistics environment.
 - Designing an integrated gripper capable of various grasping strategy for a shelf environment among logistics environments
 - Researching grasping strategy that can pick objects while avoiding external constraints in a shelf environment

GRANT

- **BK21 Research Encouragement Scholarship**  *Fall. 2024, and Spring. 2026*
- **Teaching Assistantship Scholarship** *Spring. 2022, Fall. 2022*
- **Brain Hanyang, Academic Excellence Scholarship** *Spring. 2018*